

High Quality Shape from a RGB-D Camera using Photometric Stereo



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PROBLEM STATEMENT

Depth images from RGB-D cameras:

- Noisy
- No fine details
- Missing areas



Goal: Improve the quality of depths

CONTRIBUTIONS

- Propose a novel RGB ratio model to resolve the nonlinearity and achieve similar accuracy to the previous methods.
- Introduce a robust multi-light method which outperforms the state-of-the-art approaches without any regularization term.
- First depth super-resolution method based on photometric stereo.

REFLECTANCE MODEL

Lambertian Reflectance Model



$$I = \rho S = \rho \mathbf{l}^\top \mathbf{n}$$

$\mathbf{l}$: light direction, $\mathbf{n}$: surface normal.
1st-order Spherical Harmonics (SH)

$$I = \rho(\mathbf{l}^\top \mathbf{n} + \varphi) = \rho \mathbf{s}^\top \tilde{\mathbf{n}}$$

φ : ambient light parameter. SH model accounts for 87.5% real-world illumination.

Surface Normal

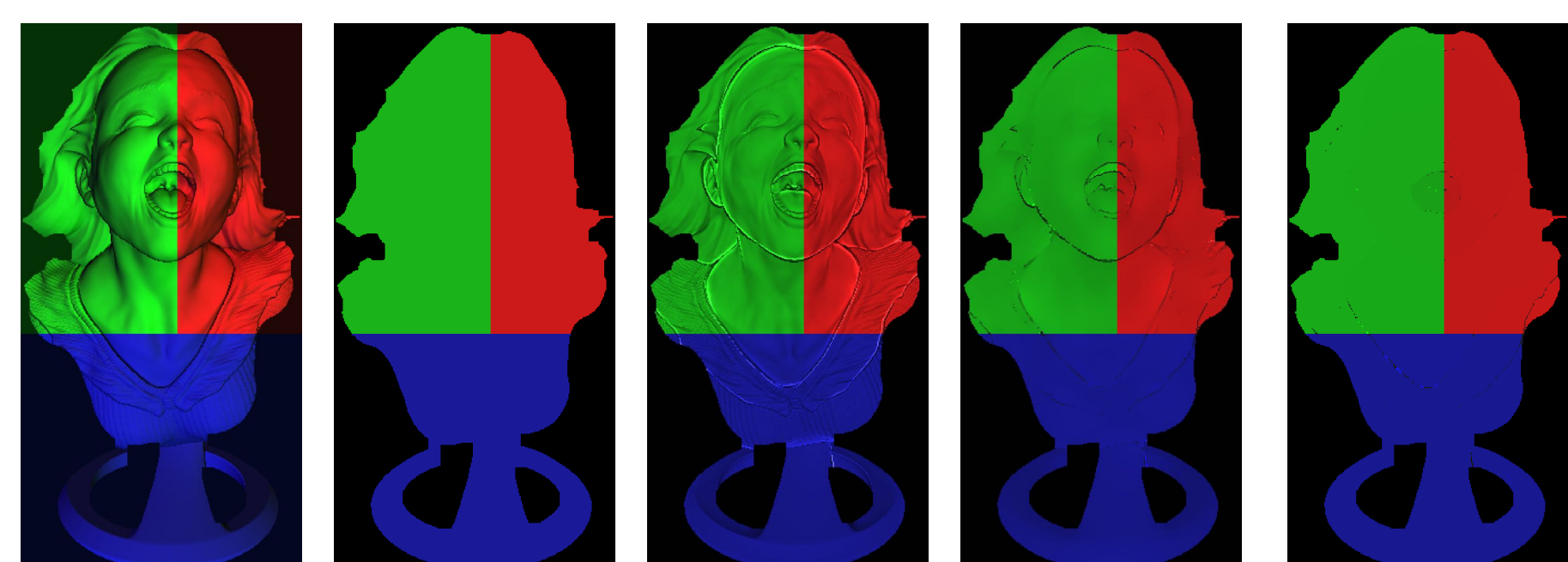
$$\mathbf{n} = \frac{1}{\sqrt{|\nabla z|^2 + 1}} \begin{pmatrix} \nabla z \\ -1 \end{pmatrix}$$

Our goal is to acquire z from the SH model.

OVERALL ENERGY

$$E(\mathbf{s}, \rho, z) = E_{\text{data}}(z) + E_{\text{shading}}(\mathbf{s}, \rho, z) + E_{\text{regu}}$$

- $E_{\text{data}}(z) = \lambda_z \|z - z_0\|_2^2$ depth data term
- $E_{\text{shading}}(\mathbf{s}, \rho, z)$ varies from methods
- E_{regu} regularization imposed on ρ or z



REFERENCES

- [1] R. Grosse *et al.* Ground Truth Dataset and Baseline Evaluations for Intrinsic Image Algorithms. In *ICCV '09*
- [2] R. Or-EI *et al.* RGBD-Fusion: Real Time High Precision Depth Recovery. In *CVPR '15*
- [3] Y. Han *et al.* High Quality Shape from a Single RGB-D Image under Uncalibrated Natural Illumination. In *ICCV '13*

RGBD-FUSION LIKE METHOD

Modification of RGBD-Fusion method [2].

$$E_{\text{shading}} = \|I - \rho \mathbf{s}^\top \tilde{\mathbf{n}}\|_2^2$$

$$E_{\text{regu}} = \lambda_\rho \|\sum_{k \in \mathcal{N}} \omega_k (\rho - \rho_k)\|_2^2 + \lambda_l \|\Delta z\|_2^2$$



RGB RATIO MODEL

Red and Green channel:

$$\frac{I_R - \rho_R \varphi_R}{I_G - \rho_G \varphi_G} = \frac{\rho_R \mathbf{l}_R^\top \mathbf{n}}{\rho_G \mathbf{l}_G^\top \mathbf{n}}$$

Nonlinearity has been resolved.

$$\rho_G(I_R - \rho_R \varphi_R) \mathbf{l}_G^\top \mathbf{n} - \rho_R(I_G - \rho_G \varphi_G) \mathbf{l}_R^\top \mathbf{n} = 0$$

$$\rho_B(I_G - \rho_G \varphi_G) \mathbf{l}_B^\top \mathbf{n} - \rho_G(I_B - \rho_B \varphi_B) \mathbf{l}_G^\top \mathbf{n} = 0$$

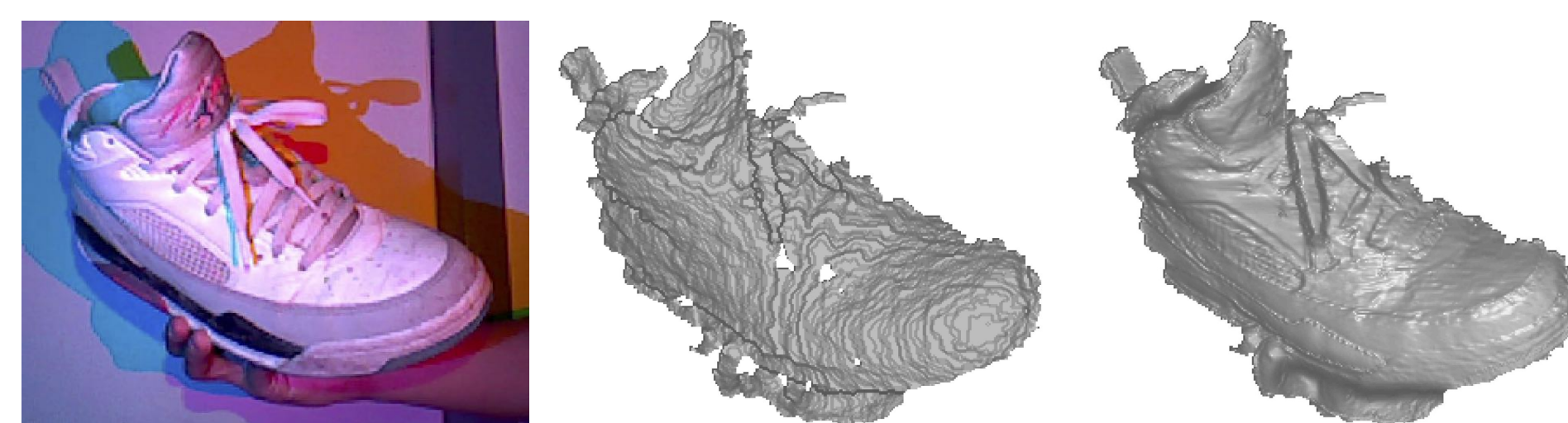
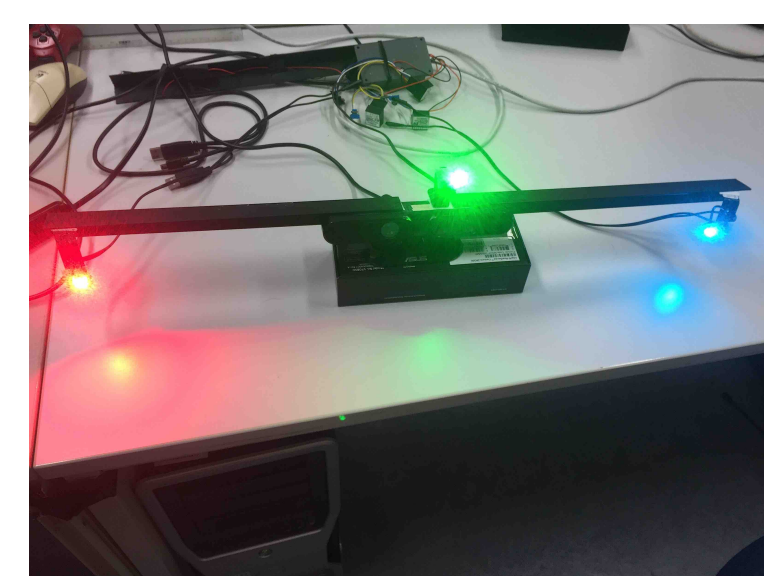
$$\rho_R(I_B - \rho_B \varphi_B) \mathbf{l}_R^\top \mathbf{n} - \rho_B(I_R - \rho_R \varphi_R) \mathbf{l}_B^\top \mathbf{n} = 0$$

$$\Rightarrow \mathcal{R}(\rho, z) = 0 \text{ (RGB ratio model)}$$

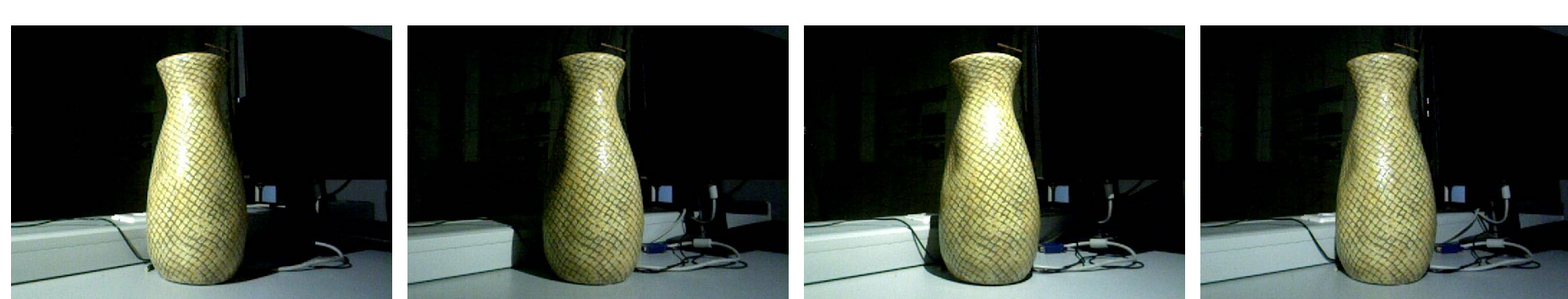
$$E_{\text{shading}} = \|\mathcal{R}(\rho, z)\|_2^2$$

$$E_{\text{regu}} = \lambda_\rho \|\omega \nabla \rho\|_2^2$$

* We add active lights to emphasize the difference among 3 channels of color images.



MULTI-LIGHT METHOD



Assuming n illumination conditions

$$E_{\text{shading}} = \sum_i^n \|I - \rho \mathbf{s}_i^\top \tilde{\mathbf{n}}\|_2^2$$

$$E_{\text{regu}} = 0$$

Depth Super-resolution

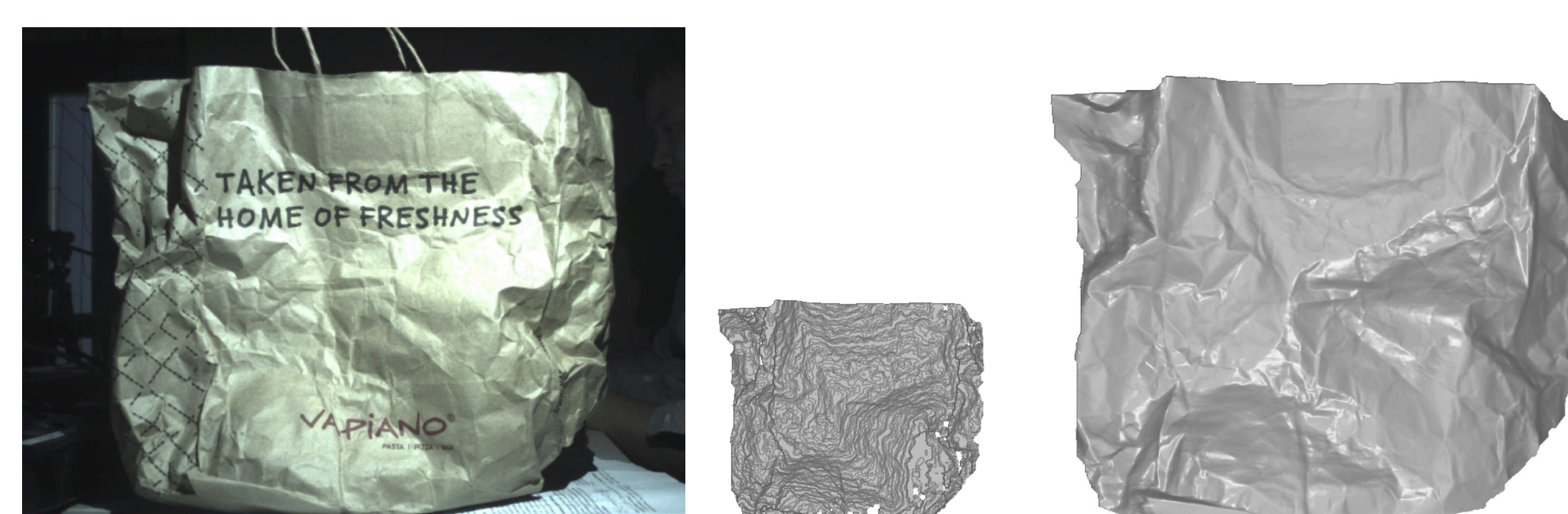
ASUS Xtion Pro Live can provide:

- 1280×960 RGB image (after cropping)
- 640×480 depth image

Goal: 1280×960 high-quality depth image

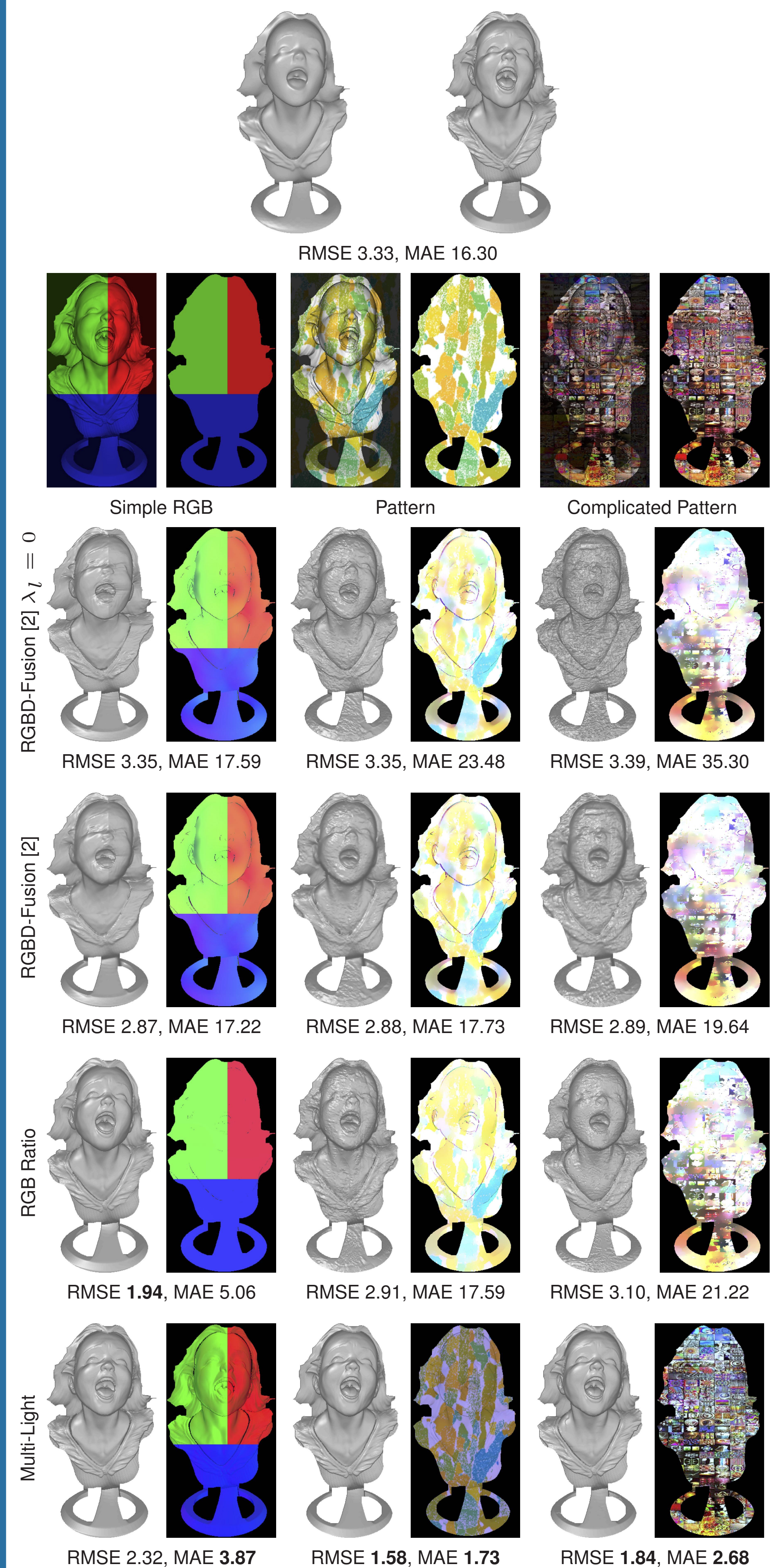
$$E_{\text{data}}(z) = \lambda_z \|Kz - z_0\|_2^2$$

K : downsampling operator.



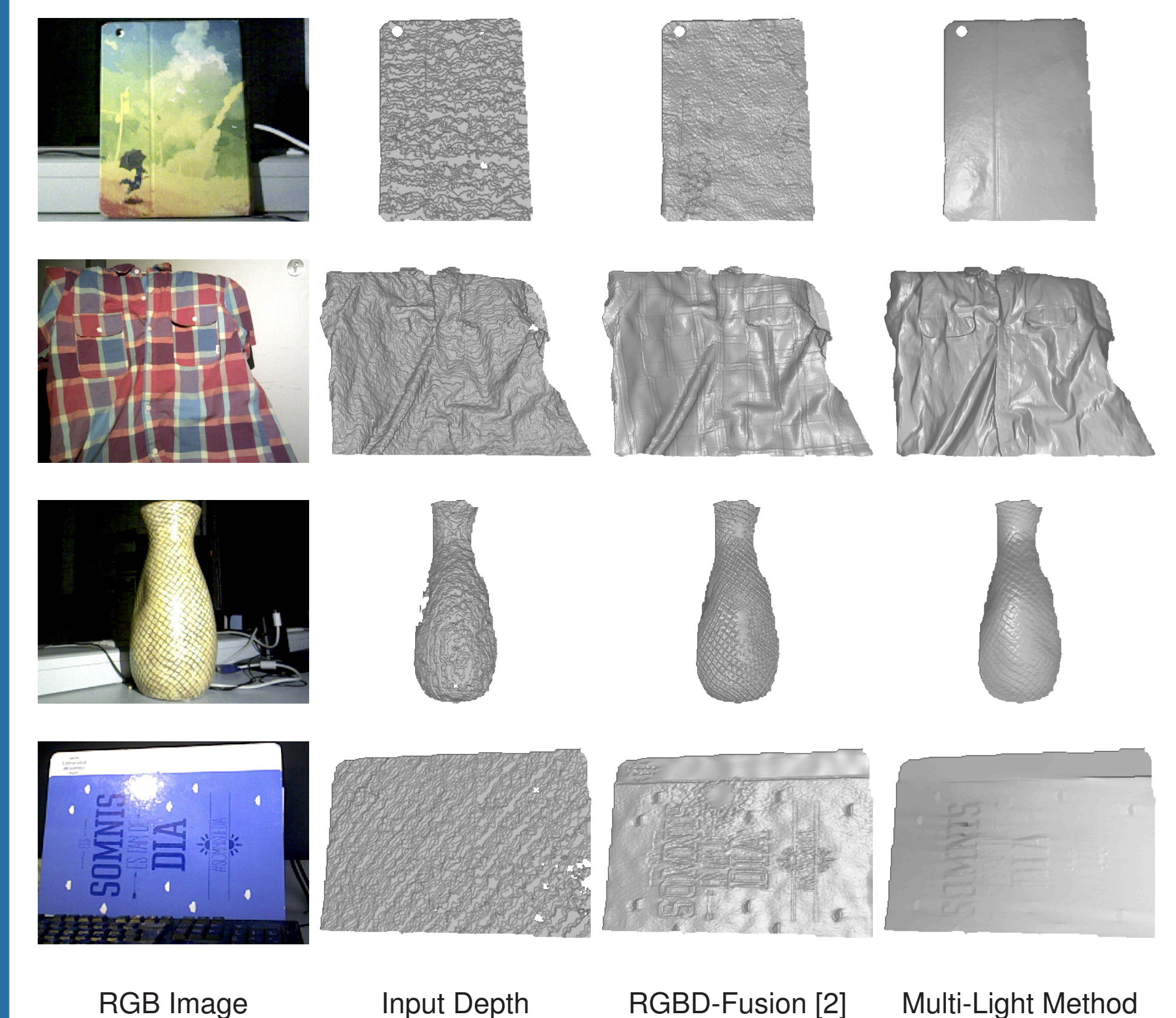
RESULTS

Synthetic Data



RMSE: root mean square error (in *mm*)
MAE: mean angular error (in degree $^\circ$)

Real Data



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